GRADUATE SCHOOL OF MANAGEMENT
UNIVERSITY OF AUCKLAND BUSINESS SCHOOL

BUSINESS ANALYTICS FOR FINTECH

BUSINESS 716

Assignment 3

March 6, 2026

Question 1 [15 points]

Overiew

Regression tree is a powerful tool used for predicting continuous outcomes by recursively splitting the data into subsets based on feature values. Each split aims to reduce the variance within the subsets, and the prediction for a new observation is made by following the path from the root to a leaf, where the predicted value is the average of the target variable in that leaf node. In this assignment, you will fit a regression tree using a dataset that records loans made on a fintech lending platform and visualize the final result.

Requirements

This is an **individual assignment**. Please complete

`BUSINFO_716_Assignment_3_Q1_Tree.ipynb`.

Detailed instructions are provided in the file.

Submission

Please save and submit your completed Jupyter notebook (`.ipynb`) file as

`Assignment_3_id`,

where `id` is your University of Auckland username. For example, if your username is "kals782", the filename should be `Assignment_3_kals782`.

Question 2: [25 points]

Overview

Understanding how to build and evaluate predictive models is a fundamental skill in data science. Regression techniques are powerful tools for uncovering relationships between variables and making data-driven decisions. In this assignment, you will enhance your ability to preprocess data, implement regression models, and interpret results—key skills that are highly valued in many industries.

Requirements

This is a group assignment. Please complete

`BUSINFO_716_Assignment_3_Q2_House_Prices.ipynb`,

ensuring that your work includes data preprocessing, model selection, performance evaluation, and interpretation of findings. Detailed instructions are provided in the file.

Submission

You do NOT need to submit the completed Jupyter notebook (`.ipynb`) file. Instead, you should work with your group members to prepare a group presentation. Each member should present one part of the data analysis. Please refer to

`BUSINFO_716_Group_Presentation_Guide.pdf`

for details.